

4(a)	straight line through the origin shows that a is proportional to x	B1
	negative gradient shows that a and x are (always) in opposite <u>directions</u>	B1
4(b)(i)	$a = -\omega^2 x$ $\omega = \sqrt{(2A / 3Y)}$	A1
4(b)(ii)	$v_0 = \omega x_0$	C1
	$= 3Y \times \sqrt{(2A / 3Y)}$ $= \sqrt{(6AY)}$	A1
4(b)(iii)	$E = \frac{1}{2} m \omega^2 x_0^2$	C1
	$= \frac{1}{2} m \times (2A / 3Y) \times (3Y)^2$ $= 3mAY$	A1
4(c)	$\omega = 2\pi / T$ $(= 2\pi / 0.75)$	C1
	$x = 1.8 \sin (8.4 t)$	A1